

Project ID :

25-26J-523

1. Topic (12 words max)

Multi-Agent Urban Traffic Signal Optimization in Simulated Environments

2. Research group the project belongs to

CoEAI - Centre of Excellence for AI

3. Specialization of the project belongs to

Software Engineering (SE)

4. If a continuation of a previous project:

Project ID	
Year	

5. Brief description of the research problem including references (200 – 500 words max) – references not included in word count.

Urban traffic congestion remains a persistent challenge for modern cities, significantly affecting commute times, fuel consumption, and air quality. In many urban areas, including cities in Sri Lanka, traffic signals operate on fixed-time or actuated logic that lacks responsiveness to real-time traffic conditions [1]. These static control systems are often unable to adapt to fluctuating vehicle volumes throughout the day or to handle non-recurring events such as accidents, roadwork, or weather disruptions [2]. This lack of adaptivity contributes to inefficient traffic flow, increased travel delays, and elevated emissions. Furthermore, emergency vehicle preemption, a critical aspect of urban traffic control is either absent or implemented using basic override mechanisms that do not consider network-wide effects, leading to safety concerns and delayed emergency response [3].

With the increasing complexity of urban mobility and the heterogeneity of traffic patterns, the limitations of traditional control methods have become more evident. The inability to dynamically respond to varying demand at intersections causes localized congestion, which often cascades into broader network-level inefficiencies [4]. In regions experiencing rapid urbanization without matching infrastructure advancement such as many parts of Sri Lanka, these issues are further exacerbated. Although a variety of microscopic traffic simulation platforms, such as SUMO, CityFlow, VISSIM, and AIMSUN, provide opportunities to test and validate adaptive traffic control strategies in virtual environments, many real-world networks remain understudied in simulated contexts due to challenges in accurate map modeling, data availability, and integration with intelligent decision-making systems [5].

In summary, the core problems identified include the inflexibility of conventional traffic signal systems to adapt to real-time and non-recurring traffic conditions, the inadequate integration of emergency vehicle routing strategies, and the broader inefficiencies that arise from these limitations. Additionally, the lack of comprehensive simulation-based studies using realistic traffic maps restricts the evaluation of intelligent control solutions in representative urban settings. These challenges are especially prominent in developing regions, where traffic infrastructure often lags growing mobility demands. Addressing these issues through intelligent, multi-agent-based traffic signal optimization could lead to more efficient, adaptive, and safer urban traffic management systems.

References

- [1] Skoropad, P., et al. (2025). *City-Wide Traffic Signal Timing Optimization Based on Dynamic Traffic Patterns*. Sustainability, 17(8), 3383. <https://www.mdpi.com/2071-1050/17/8/3383>
- [2] Tang, Y., & Duan, J. (2024). *Traffic Signal Control Based on Spatio-Temporal Graph Learning*. Actuators, 13(7), 251. <https://www.mdpi.com/2076-0825/13/7/251>
- [3] Su, Y., et al. (2023). *Multi-Objective Emergency Vehicle Priority Control in Mixed Traffic*. Transportation Research Part C: Emerging Technologies, 146, 103955. <https://arxiv.org/abs/2206.13441>
- [4] Abdoos, M., et al. (2011). *Traffic Light Control in Non-Stationary Environments Based on Multi Agent Q-Learning*. In Proceedings of the 14th IEEE ITSC. <https://ieeexplore.ieee.org/document/6083133>
- [5] López, P. A., et al. (2018). *Microscopic Traffic Simulation Using SUMO*. In Proceedings of the 21st IEEE International Conference on Intelligent Transportation Systems (ITSC). <https://ieeexplore.ieee.org/document/8569938>

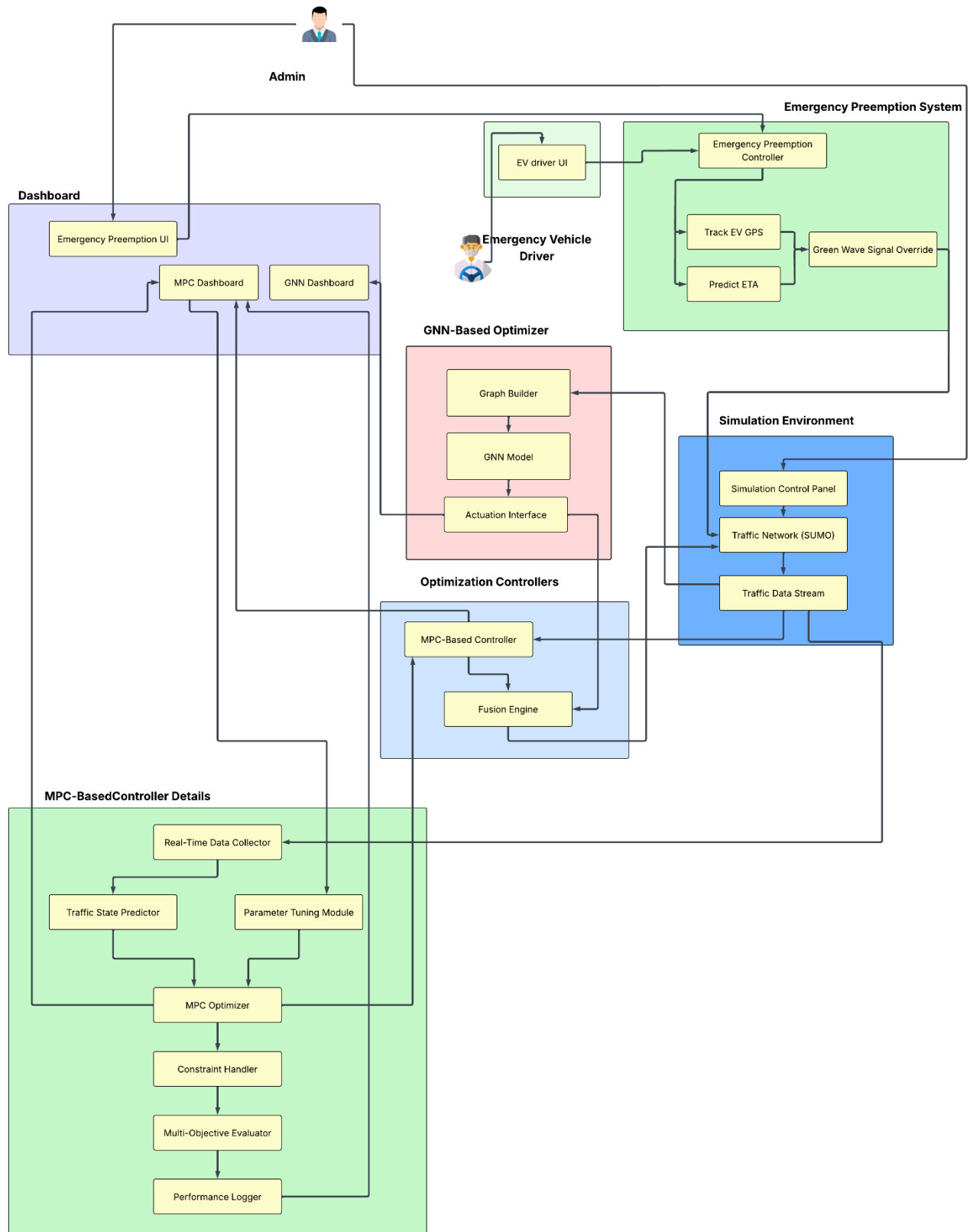
6. Brief description of the nature of the solution including a conceptual diagram (250 words max)

To address the limitations of existing traffic control systems, this project proposes the development of a modular, city-scale traffic signal optimization framework. The solution integrates cutting-edge AI and control methodologies to build a system that is both intelligent and adaptable to real-time urban traffic conditions.

The system consists of four main components. First, a simulation environment will be built using SUMO to provide a realistic and configurable city traffic model. Second, a Graph Neural Network (GNN) module will be developed to optimize traffic signal timings across intersections based on the spatial and temporal relationships of traffic flows. This will enable adaptive and coordinated signal control to reduce congestion and improve traffic throughput.

Third, a Model Predictive Control (MPC) module will be implemented as an alternative strategy that allows real-time computation of optimal signal phases over a prediction horizon, providing flexibility and robustness under constraints. Finally, an emergency vehicle preemption system will be designed to detect emergency routing needs and dynamically adjust traffic signals to ensure safe and fast passage.

Each module will be supported by interactive dashboards for visualization and control, allowing users to monitor system behavior and compare performance metrics. The integrated solution aims to significantly reduce traffic congestion, improve emergency response times, and provide a scalable prototype suitable for future deployment in real-world smart city environments.



[Link to the diagram](#)

7. Brief description of specialized domain expertise, knowledge, and data requirements (300 words max)

This project spans AI, optimization, transportation systems, and software engineering. Key domain areas include:

- Graph Machine Learning: Knowledge of graph theory, feature encoding, and training GNNs using libraries like PyTorch Geometric.
- Control Theory: Understanding of MPC principles, traffic flow modeling, and constrained optimization.
- Traffic Simulation: Experience with microscopic traffic simulators (e.g., SUMO, CityFlow, VISSIM, AIMSUN) to model realistic urban traffic environments.
- Emergency Systems: Familiarity with pathfinding, route prediction, and signal preemption strategies.
- Full Stack Development: Building interactive web dashboards using React/Node.js for simulation control and visualization.

Data Requirements:

- Synthetic traffic flow data generated using traffic simulation platforms.
- Real-world traffic flow patterns for model validation.
- Manually defined emergency vehicle routes for preemption testing.

All components are modular and software-based, making them easy to test and integrate.

8. Objectives and Novelty

Main Objective Develop a modular, AI-driven traffic signal optimization system for urban networks, using GNN and MPC, with integrated emergency preemption and real-time simulation dashboards.			
Member Name with Registration No	Sub Objective	Tasks	Novelty
Sandaruwan G. D. I. IT22566898	Develop a realistic city-scale traffic simulation and control panel	<u>SE</u> - Design and implement integration architecture for selected simulation platform. - Develop RESTful APIs for simulation control and real-time data exchange. - Create web-based control panel using modern frameworks. <u>Traffic Simulation Related Tasks</u> - Conduct comparative analysis of existing traffic simulators using defined criteria. - Configure and set up selected simulation platform for urban traffic scenarios. - Implement scenario management system for	- Innovative real-time parameter adjustment and scenario switching system with rollback capabilities, addressing limitations of existing static simulation approaches. - Standardized interface protocol for simultaneous integration of multiple AI approaches (GNN, MPC) with traffic simulators, enabling comparative analysis and hybrid control strategies. - Comprehensive web-based monitoring system combining simulation data with AI decision insights, providing multi-layered

		dynamic traffic condition switching. Develop real-time monitoring and visualization of traffic simulation data.	real-time traffic analysis capabilities
Kahawaththa K. A. O. N. S. IT22591302	Implement a GNN-based adaptive traffic signal optimizer	<u>SE</u> - Develop a comprehensive dashboard for real-time visualization and analysis of traffic forecasts - Implement a mobile interface for public users to query filtered traffic data (e.g., current delays, predicted congestion, travel times) <u>AI/ML</u> - Design a Graph Neural Network that takes Node features and the edge features to optimize signal timings - Refine the GNN via expert-guided imitation to minimize delay (and optionally fuel or emissions).	- Decoupled Real-Time Dashboard for live KPI visualization and manual override, enabling human-in-the-loop control - Mobile interface that lets public users query real-time and forecasted traffic data (e.g. delays, congestion levels, travel times). - A unified controller that handles all types of road elements (intersections, crosswalks, bus lanes) in one model

		- Add real-time confidence checks that trigger a safe fallback to the expert when uncertainty is high. <u>Usage of Simulation</u> - Run the simulator step by step, read the live traffic data, and send the GNN's signals back into it. - Create various traffic scenarios to record state–action examples for improving the model.	
Jayakody R. H. IT22598950	Build an MPC-based signal controller for performance comparison	<u>SE</u> - Develop a comprehensive dashboard for real-time visualization and analysis of traffic forecasts - Design the system to allow real-time updates and fine-tuning of traffic objectives and constraints <u>AI/ML</u> - Use Model Predictive Control (MPC) to forecast traffic flow patterns and compute optimal signal plans that reduce congestion and vehicle delay. - Formulate traffic signal optimization as a constrained decision-making problem with defined objectives and rules.	- Interactive dashboard allows live tuning of prediction horizon, cost priorities (e.g., delay vs. queue length), and signal constraints. - Enables real-time visualization of traffic signal decisions and predictions on an interactive map-based interface, allowing users to easily monitor current light statuses, predicted traffic flow, and system behavior through mobile interface.

		Implement cost functions that prioritize smoother traffic flow while respecting minimum/maximum signal durations and junction safety	
Randika D. G. T. IT22598714	Implement an emergency vehicle preemption system	<u>SE</u> - Build a mobile app for EV drivers to route selection, view ETA, route, and signal status - Implement a dynamic green-wave controller to intelligently override traffic signals for the emergency vehicle <u>AI/ML</u> - Develop an AI-powered prediction system to forecast vehicle arrival times (ETA) for upcoming junctions - Create an AI model for conflict minimization to ensure safe preemption decisions <u>Usage of Simulation</u> - Get GPS-like EV tracking data from the simulation - Determine the most efficient routes for EVs by analyzing real-time traffic flows from the simulation	- Mobile Driver Interface for Emergency Vehicles - A predictive model to forecast vehicle arrival times - Conflict minimization for safer, smarter preemption decisions

9. Individual component description of how it is complied with the specialization.

Member Name with Registration No	Description
Sandaruwan G. D. I. IT22566898	Simulation Environment & Control Panel This component involves setting up and managing a real-time, large-scale traffic simulation using traffic simulation platforms, and developing a web-based control panel for user interaction. These tasks require software engineering skills such as full-stack development, real-time data processing, and system integration. The component strongly aligns with Software Engineering specialization through its focus on system design, user interface creation, and simulation control logic.
Kahawaththa K. A. O. N. S. IT22591302	GNN-Based Traffic Signal Optimizer This component applies Graph Neural Networks to dynamically control traffic signals based on real-time data. The implementation includes training, integrating, and testing AI models, as well as visualizing results through a web dashboard. It aligns with the Software Engineering specialization through its emphasis on algorithm implementation, software architecture, and dashboard development using libraries and frameworks.
Jayakody R. H. IT22598950	MPC-Based Controller & Visualization Dashboard This module involves developing a Model Predictive Control system for optimizing traffic light timing and creating a dashboard to visualize predictive outcomes. Tasks require advanced programming, mathematical modeling, and UI development. It fits within the Software Engineering specialization by combining software development practices with predictive algorithm integration and system feedback visualization.
Randika D. G. T. IT22598714	Emergency Vehicle Preemption System This component simulates emergency vehicle routes and overrides standard traffic logic to ensure fast response times. It includes real-time system responses, a mobile app for emergency drivers, machine learning-based ETA prediction, and signal control integration. These tasks are closely aligned with the Software Engineering specialization due to the focus on event-driven systems, API integration, and responsive user interface development.

10. Supervisor details

	Title	First Name	Last Name	Signature
Supervisor				
Co-Supervisor				
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				

This part is to be filled by the Topic Screening Staff members.

- a) Does the chosen research topic possess a comprehensive scope suitable for a final-year project?

Yes		No	
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- b) Does the proposed topic exhibit novelty?

Yes		No	
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- c) Do you believe they have the capability to successfully execute the proposed project?

Yes		No	
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- d) Do the proposed sub-objectives reflect the students' areas of specialization?

Yes		No	
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- e) Supervisor's Evaluation and Recommendation for the Research topic:

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Acceptable: Mark/Select as necessary

Topic Assessment Accepted	
Topic Assessment Accepted with minor changes*	
Topic Assessment to be Resubmitted with major changes*	
Topic Assessment Rejected. Topic must be changed	

* Detailed comments given below

Comments

Staff Member's Name	Signature

***Important:**

1. According to the comments given by the evaluator, make the necessary modifications and get the approval by the **Evaluator**.
2. If the project topic is rejected, identify a new topic, and request the RP Team for a new topic assessment.